

Toxic cultures: E-cigarettes and the oral microbial exposome

Prem Prashant Chaudhary , PhD

Staff Scientist
Epithelial Therapeutics Unit
Laboratory of Clinical Immunology and Microbiology
National Institute of Allergy and Infectious Diseases
National Institutes of Health



E-cigarette prevalence in the U.S.

In 2021

4.5%

Of adults

In 2022

14.1%

Of high-schoolers

3.3%

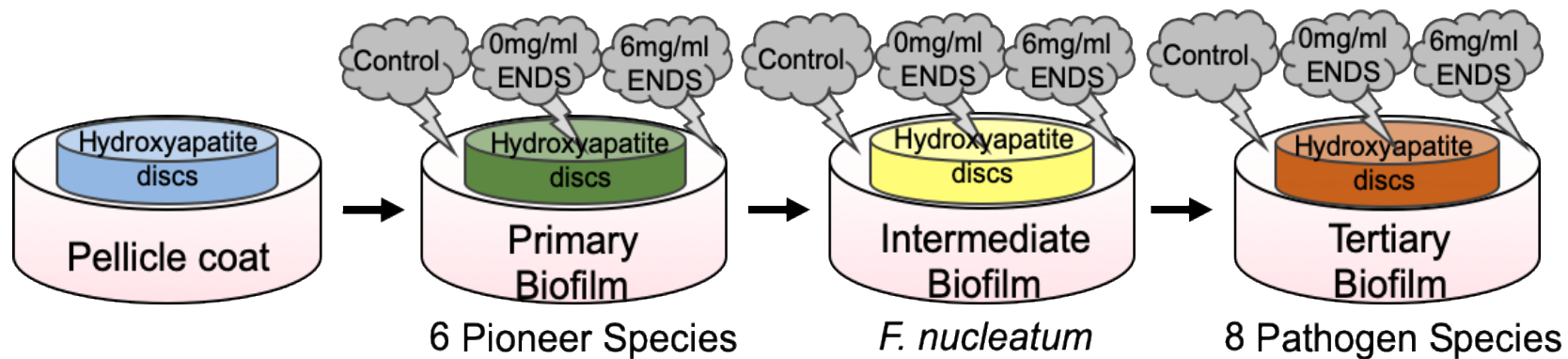
Of middle-schoolers

Reported e-cig use^{1,2}

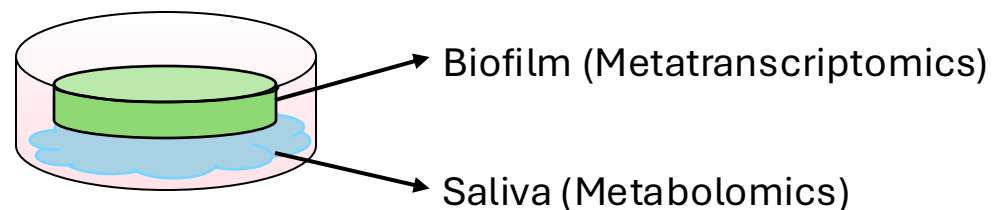
1. Cornelius ME, Loretan CG, Jamal A, et al. Tobacco Product Use Among Adults – United States, 2021. MMWR Morb Mortal Wkly Rep 2023;72:475–483. DOI: <http://dx.doi.org/10.15585/mmwr.mm7218a1>
2. Cooper M, Park-Lee E, Ren C, Cornelius M, Jamal A, Cullen KA. *Notes from the Field*: E-cigarette Use Among Middle and High School Students — United States, 2022. MMWR Morb Mortal Wkly Rep 2022;71:1283–1285. DOI: <http://dx.doi.org/10.15585/mmwr.mm7140a3>

Methods

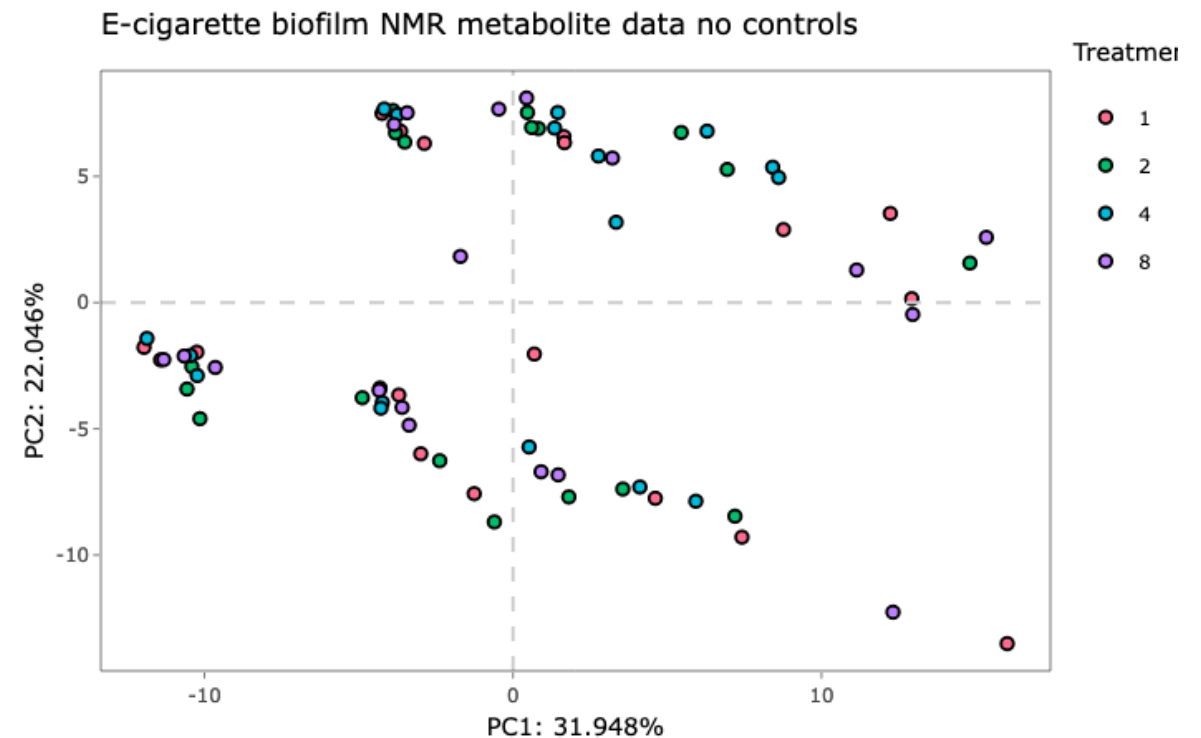
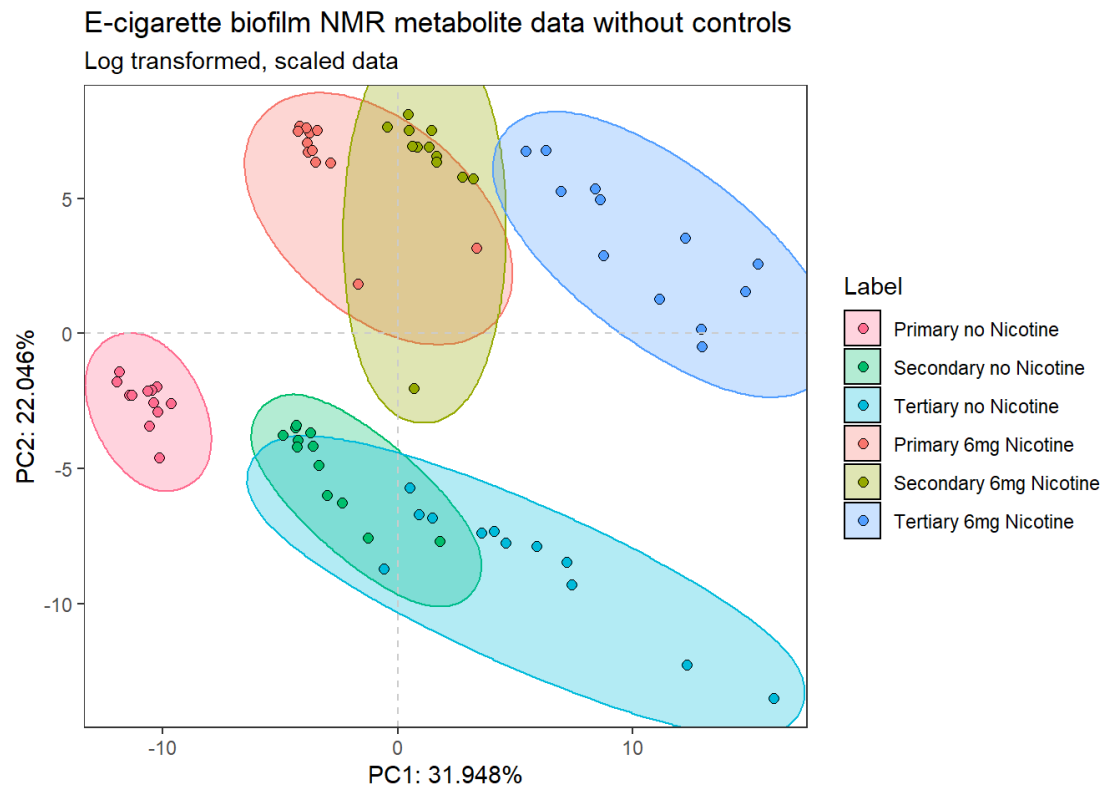
- 3 exposure conditions:
 - Unflavored ENDS aerosol (6mg/ml nicotine)
 - Unflavored nicotine-free ENDS aerosol
 - Clean air controls



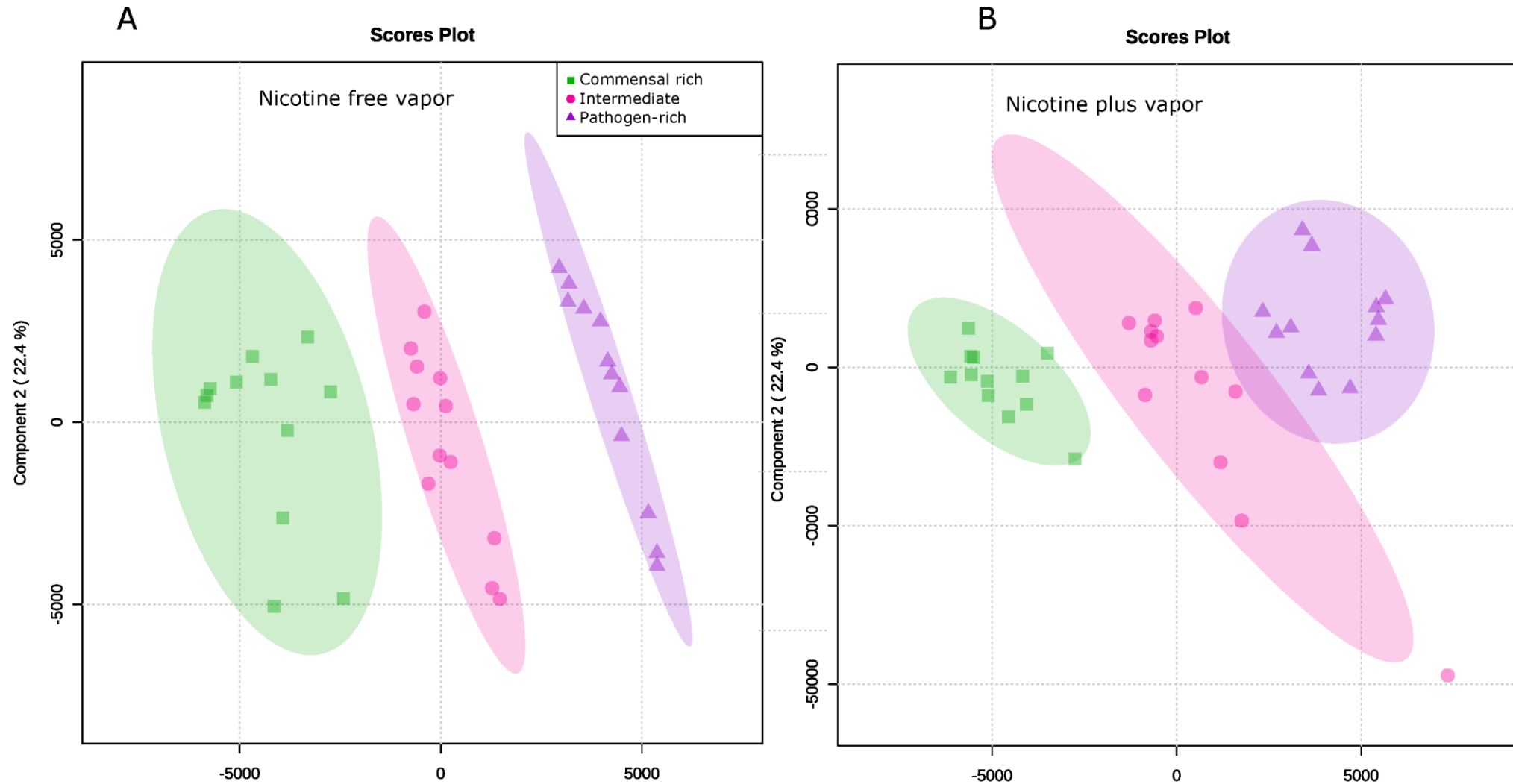
1 hour
2 hours
4 hours
8 hours



E-cigarette aerosol metabolism occurs in the first hour of exposure and is differentially impacted by microbial composition and nicotine



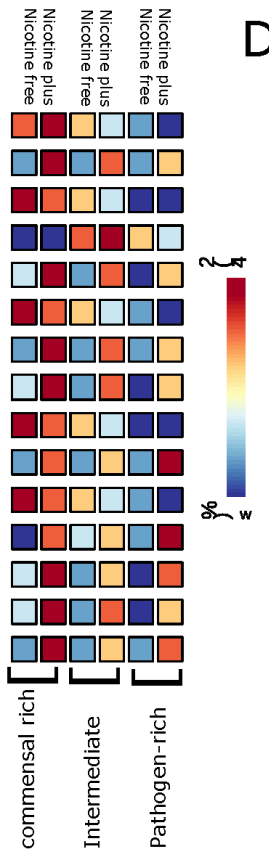
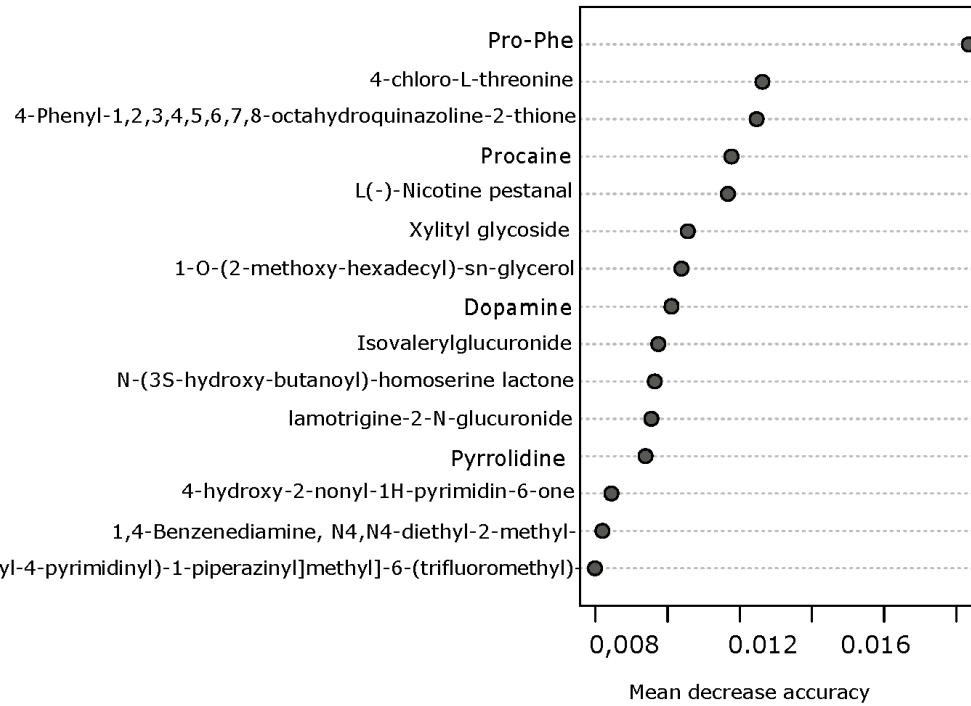
Microbial composition is a determinant of metabolite profile



Both microbial composition and biofilm composition drive metabolomic profile

C

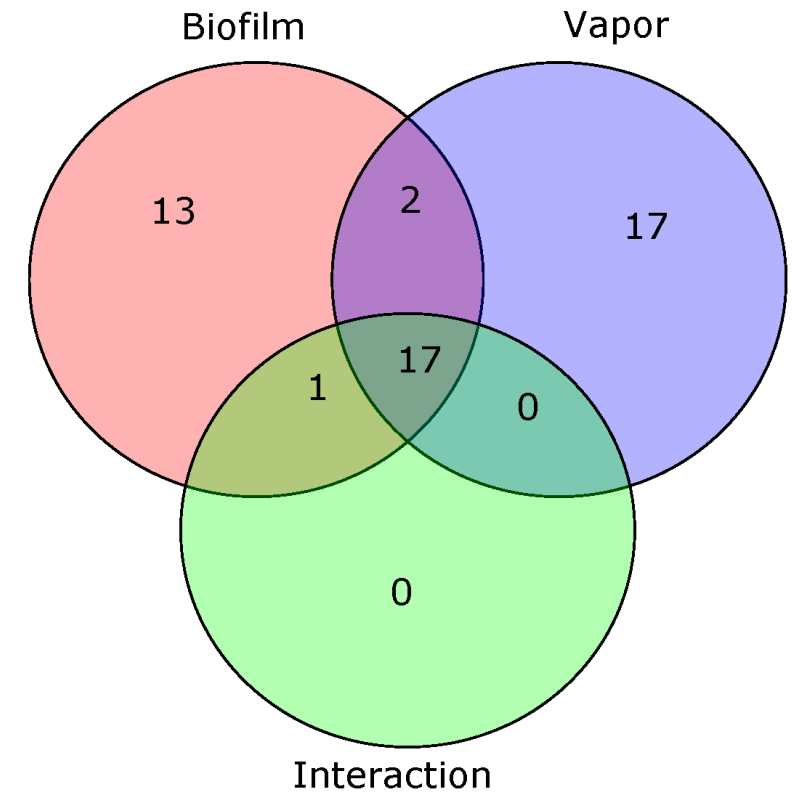
Component 1 (39.1 %)



D

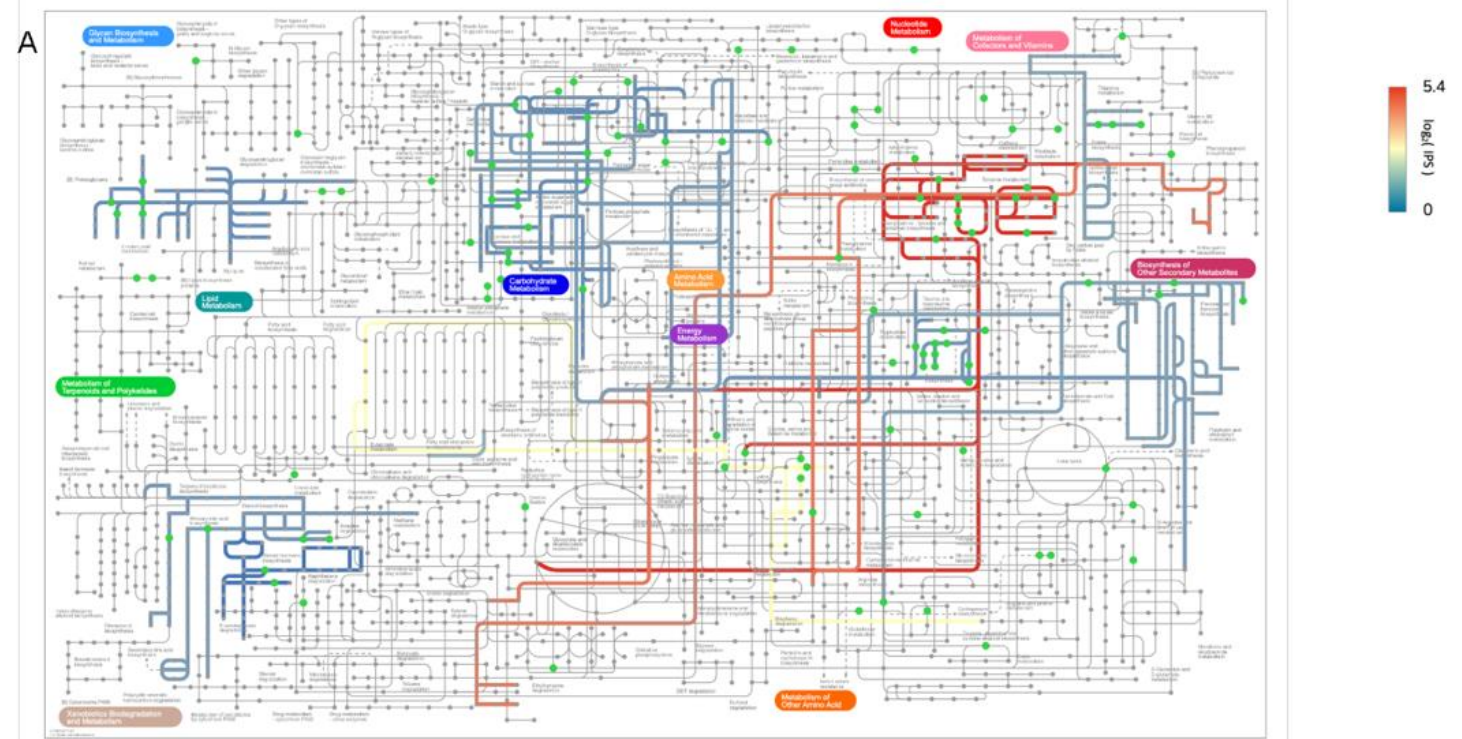
Component 1 (39.%) P

P<0.05, ANOVA



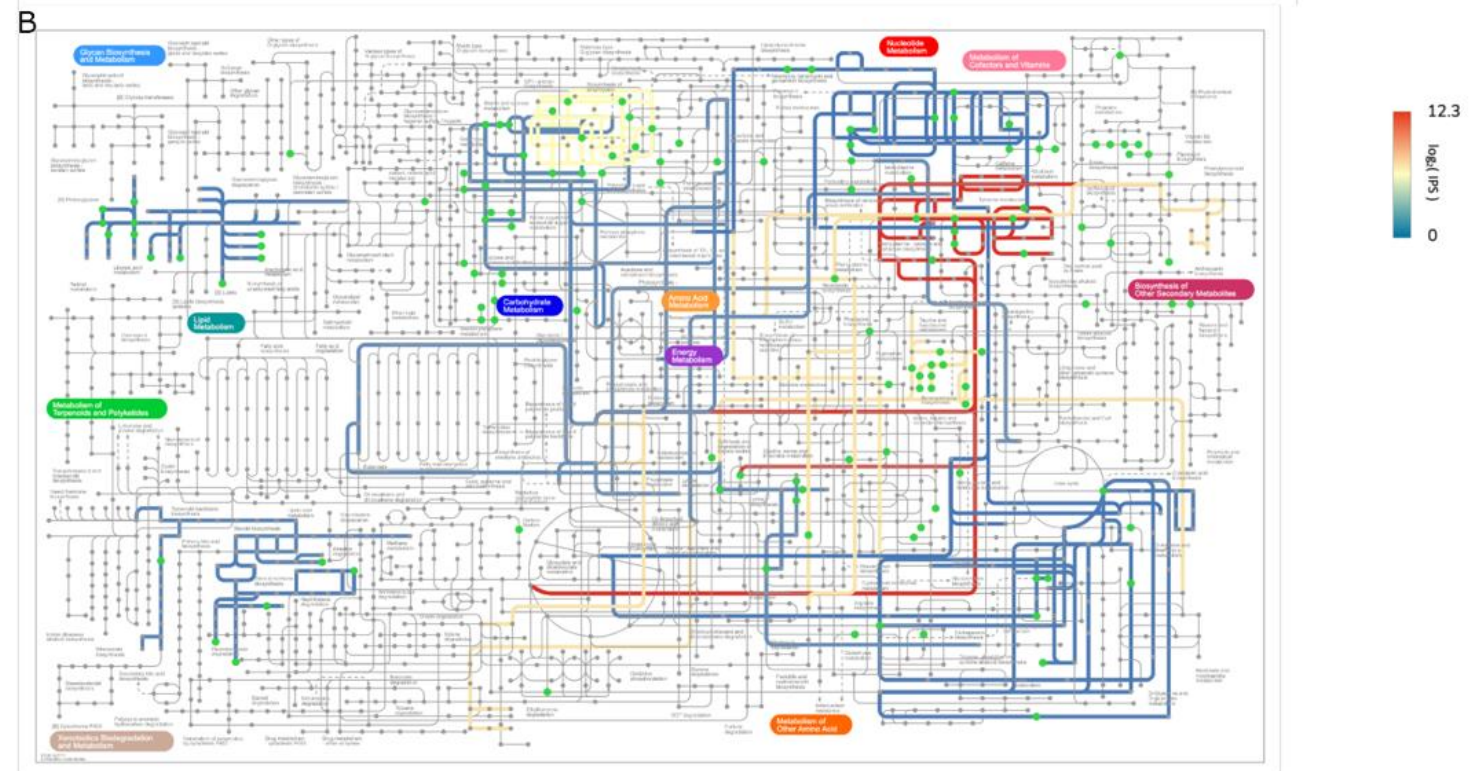
Commensal-rich biofilm pathway enrichment

- Pathways enriched:
 - Lipid, carbohydrate, and energy metabolism
 - Xenobiotic degradation
 - Intermediate metabolite generation



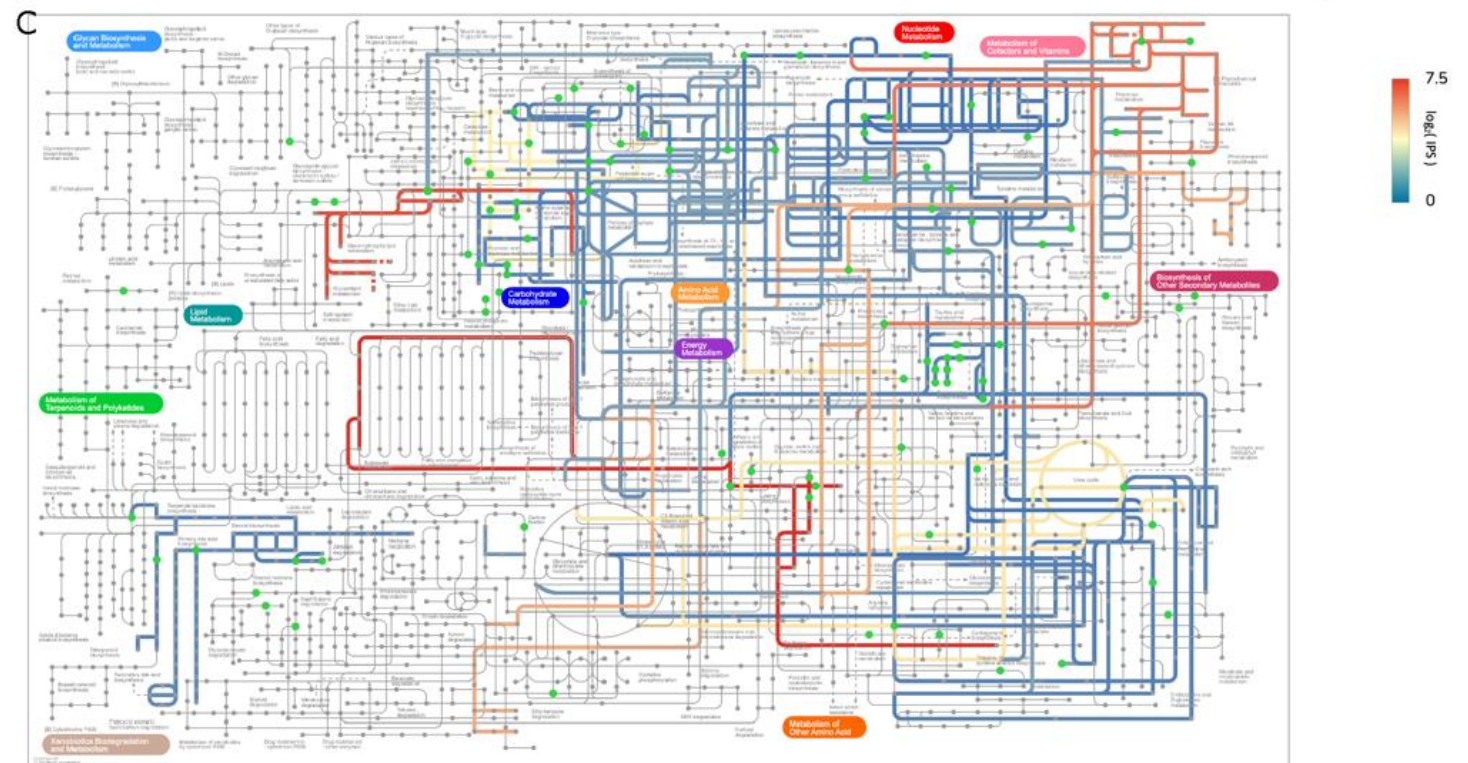
Intermediate biofilm pathway enrichment

- Pathways enriched:
 - Nucleotide synthesis
 - Beta-alanine and alanine
 - Riboflavin and tyrosine
 - Vancomycin
 - Carbohydrate metabolism
 - Steroid biosynthesis



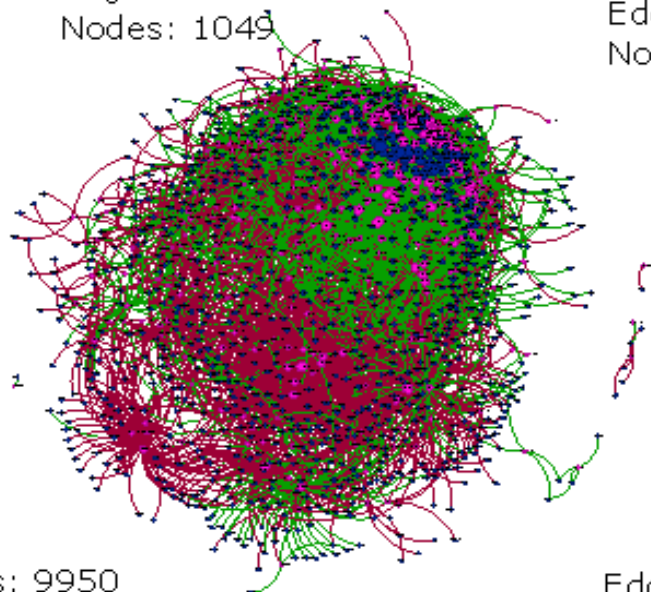
Pathogen-rich biofilm pathway enrichment

- Pathways enriched:
 - Biosynthesis of flavonoids and indoles
 - Ascorbate and alkaloid synthesis
 - Steroid biosynthesis
 - Glycan biosynthesis and degradation

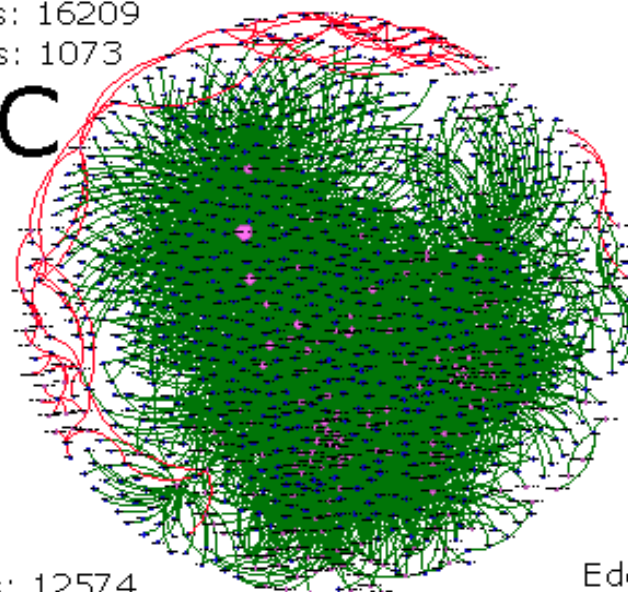


B

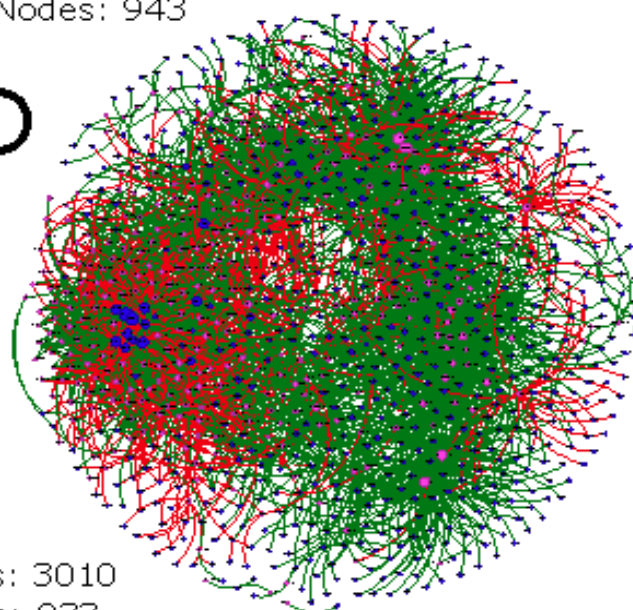
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Nodes: 1049

**C**

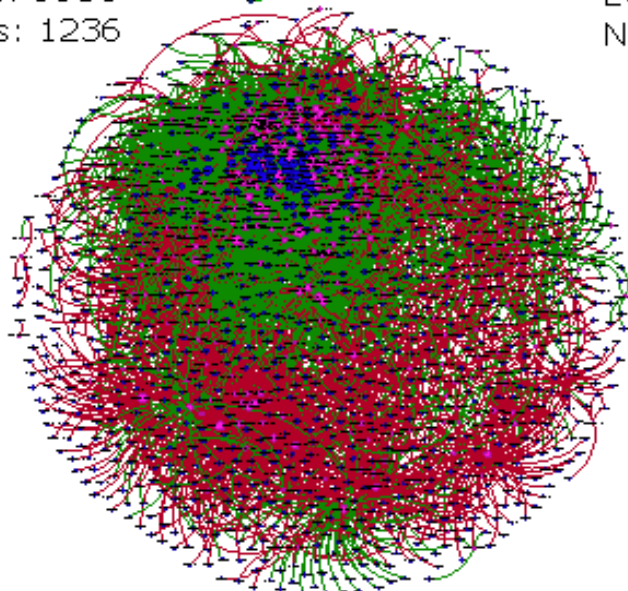
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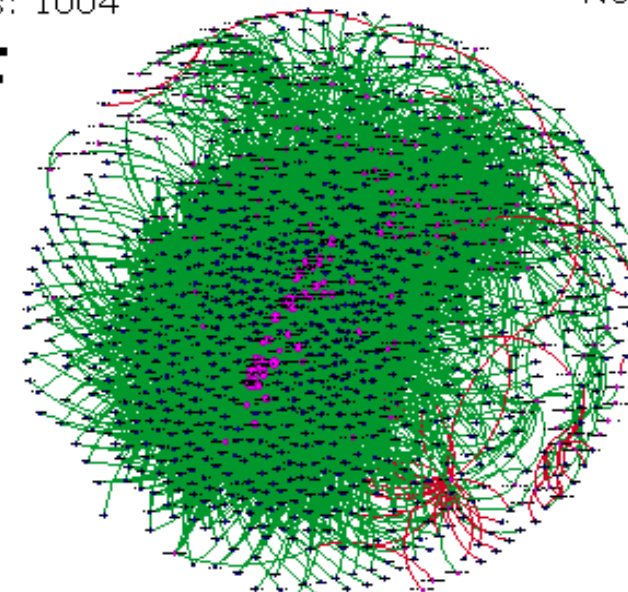
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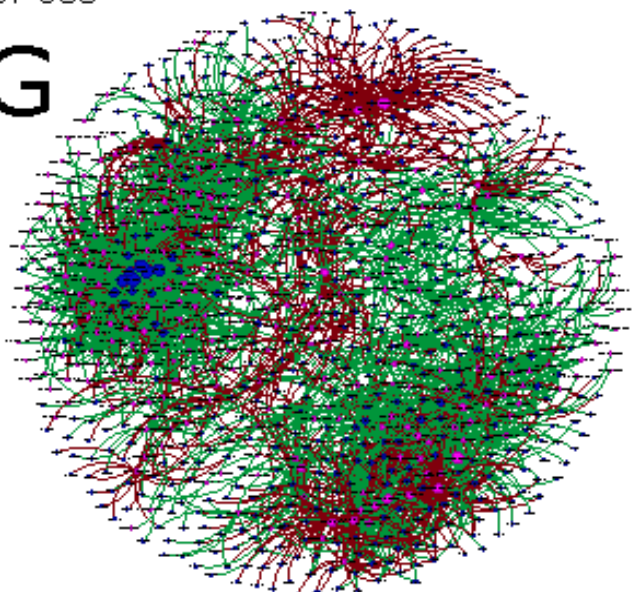
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**F**

Edges: 12574
Nodes: 1004

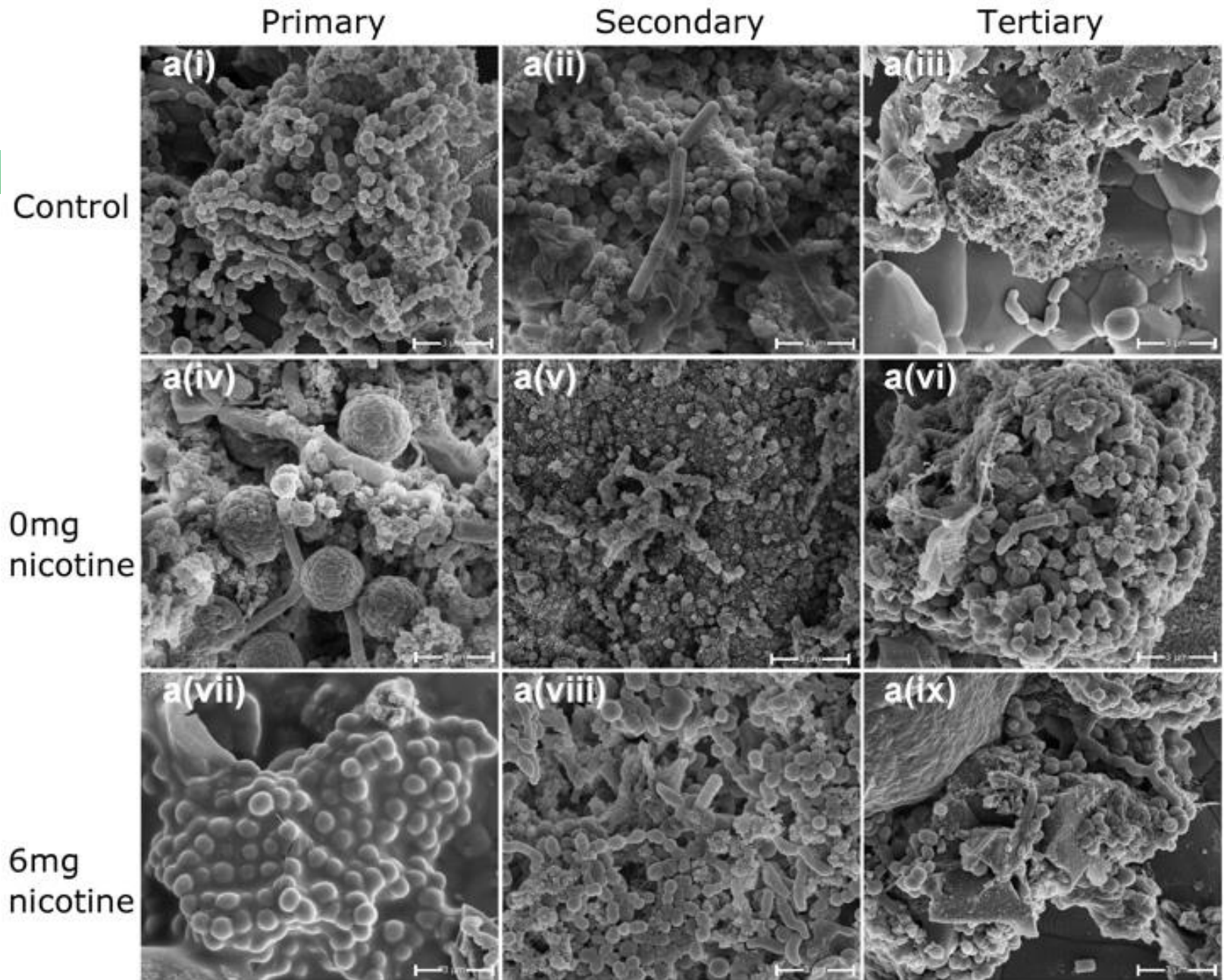
**G**

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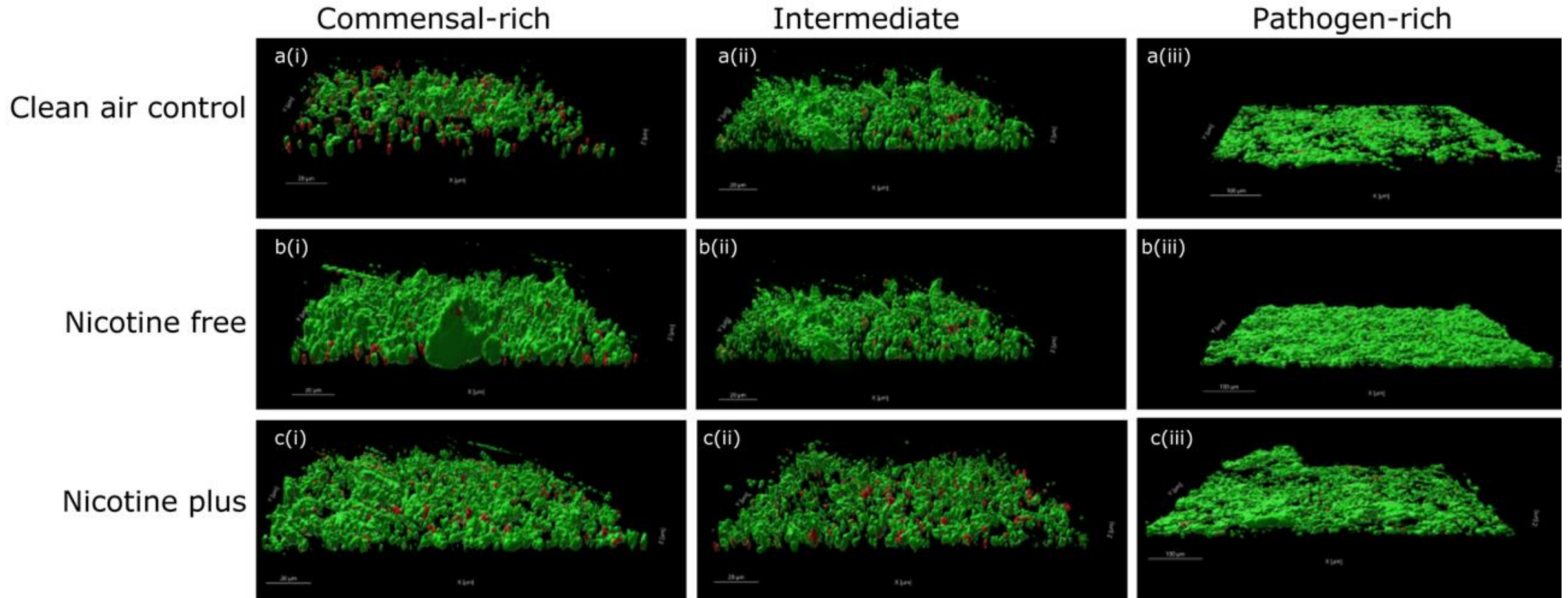


SEM imaging

E-cigarette exposure
changes biofilm
structure

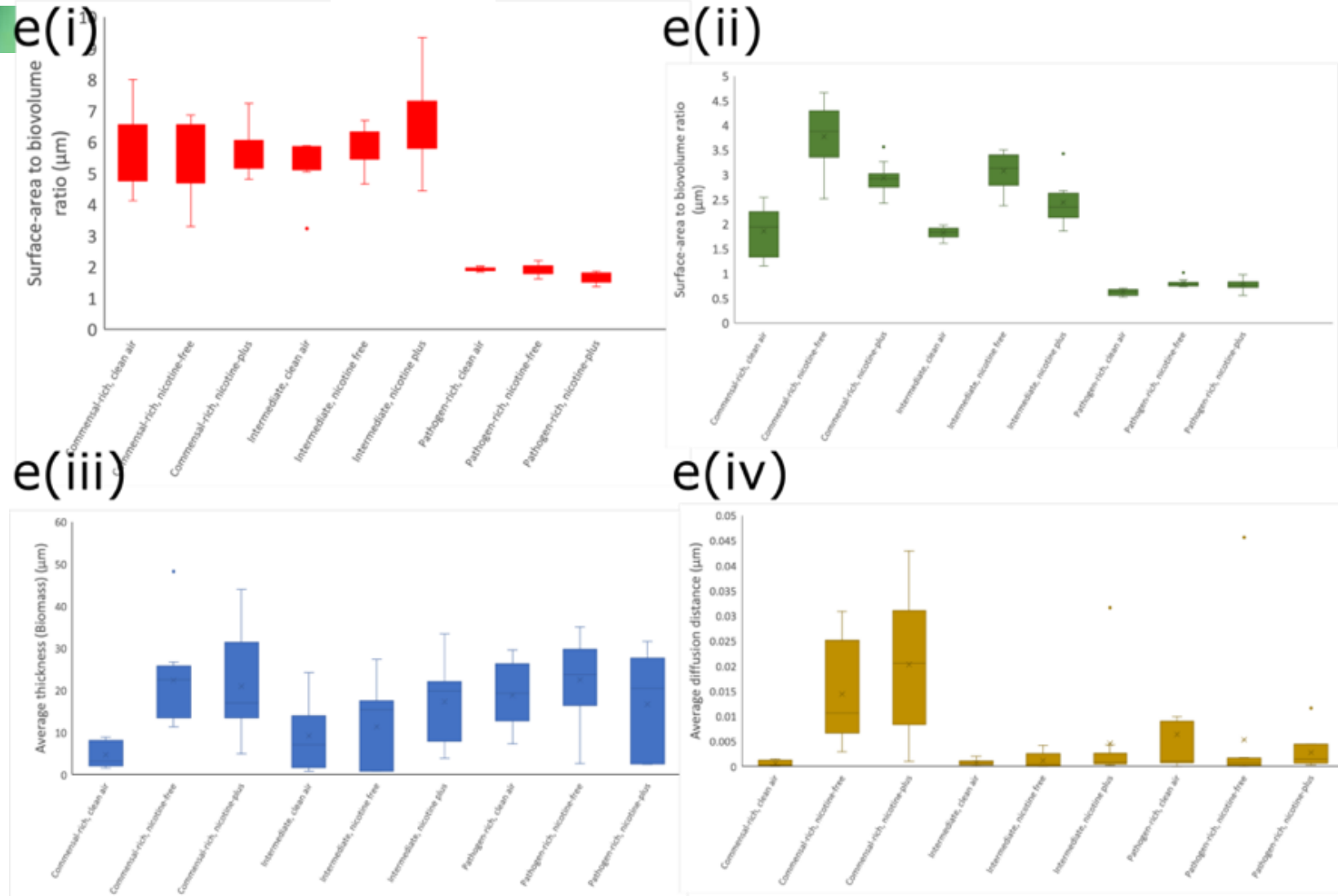


CLSM imaging of live stained biofilms



E-cigarette exposure alters biofilm topography

- Higher SA / Vol ratio in e-cig exposed commensal-rich biofilms



P<0.001, Dunn's test with joint ranks

Summary



- Oral biofilms are capable of metabolizing e-cigarette aerosol
 - Occurs within first hour of exposure
- Bacterial transcription is significantly altered in an e-cigarette rich environment
 - Nicotine further alters transcription
- Annotated metabolites and transcriptional pathways point to antibacterial agents playing a key role in why biofilms are impacted by e-cigarette aerosol
- Biofilm structure is significantly altered, likely via quorum sensing and cell wall, capsule, and ECM synthesis

Acknowledgements



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